Ratnadeep Bhattacharyya

Product Engineer – AI Systems, Backend Platforms & Distributed Infrastructure

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Summary

Senior Software Engineer with 7+ years of experience building distributed backend systems, production AI/LLM applications, and cloud-native platforms. Experienced in designing agentic workflows, RAG and retrieval systems, asynchronous processing, high-throughput APIs, and reliable distributed services using Go, Python, TypeScript, Kubernetes, and AWS/GCP. Built systems supporting 1M+ daily API requests and production AI workflows serving thousands of users. Strong track record of owning products end-to-end, from ambiguous requirements and system design through implementation, deployment, observability, and production iteration.

Experience

Springboard (SlideRule Inc.) India (Remote)

Software Engineer II (Fullstack) Jan 2025 - Present

- Built an **agentic AI grading system** using LLM orchestration, contextual retrieval, and asynchronous processing to evaluate submissions, rubrics, and student context, reducing grading turnaround from **6 days to 2 hours with 93% accuracy across 5,000 students**.
- Built and owned backend services for Springboard's mentorship platform using **Python, Go, React, and MySQL**, automating student–mentor scheduling based on availability and reducing manual coordination.
- Built **AI call-summary workflows** that convert mentor-session transcripts into structured summaries, action items, next steps, and support-risk indicators, reducing manual follow-up for student support teams.
- Designed **distributed workflow orchestration** with retries, idempotency controls, fault tolerance, and operational traceability across asynchronous grading, mentorship, and support workflows.
- Built observability tooling using **Prometheus, Grafana, structured logging, distributed tracing, and New Relic** to improve debugging, alerting, and production visibility.
- Deployed and operated workflow services on **Kubernetes (GKE)** with CI/CD pipelines, improving scalability, release reliability, and production availability.

TarkaLabs India (Remote)

Technical Consultant Jan 2024 - Nov 2024

- Worked directly with stakeholders and end users to understand workflows, retrieval needs, and customer-specific requirements for a **self-hosted agentic RAG platform**.
- Built and delivered a self-hosted **agentic RAG system** using Python, FastAPI, PostgreSQL, Qdrant, LangChain, and LlamaIndex for querying structured and unstructured enterprise knowledge.
- Designed retrieval and reasoning workflows using **vector search, hybrid search, reranking, prompt rewriting, contextual retrieval, NL-to-SQL, and NL-to-API** to improve relevance and automate customer-specific workflows.
- Implemented **RAGAS-based evaluation, PII filtering, prompt safety checks, output validation, and observability** to improve reliability and production quality across AI workflows.
- Built scalable **FastAPI backend services** for asynchronous ingestion, retrieval orchestration, external API integrations, and agentic workflow execution.
- Owned multimodal ingestion for **PDFs, Word documents, Excel files, images, tables, audio, and video transcripts**, enabling enterprise content to be processed for downstream RAG workflows.
- Developed custom chunking and metadata strategies to preserve **page references, table context, and source traceability**, improving citation quality and generated-answer trust.

Nagarro India (Remote)

Senior Software Engineer July 2023 - Dec 2023

- Improved reliability across gift and payment workflows handling **1M+ daily API requests** by redesigning service communication patterns, reducing cascading failures, and improving fault isolation.
- Built and optimized a high-volume gift microservice using **Node.js, NestJS, MySQL, RabbitMQ, and Redis**, integrated with payment and order services.
- Applied **Saga-based distributed transaction patterns** with RabbitMQ-based asynchronous processing to improve consistency, recovery, and failure handling across multi-service workflows.
- Enhanced backend performance through **MySQL query optimization, Redis caching, and asynchronous processing** to support low-latency transactions under high traffic.
- Designed reliable purchase workflows coordinating gift issuance, payments, order state, and dependent microservices.
- Strengthened production debugging and operational visibility through structured logging and monitoring.

Dentsu World Services India (Remote)

Software Engineer July 2021 - June 2023

- Built **Python and Go microservices** for a multi-tenant analytics platform serving **6K+ daily users and 200K+ daily API requests** across telecom, manufacturing, and logistics customers.
- Built high-throughput **Go-based event ingestion systems** for distributed customer data streams used in analytics, reporting, and operational workflows.
- Designed **event-driven architectures and asynchronous pipelines** to improve scalability, fault tolerance, and service communication.
- Developed AWS-based backend services using **ECS/EKS, Lambda, S3, and Airflow** for batch, near-real-time, and distributed processing workloads.
- Designed external-facing APIs and webhook workflows allowing third-party clients to submit analytics jobs, consume outputs, and receive asynchronous completion notifications.
- Designed ingestion and storage workflows using **MongoDB and Snowflake** for high-volume analytics processing.
- Improved throughput, cost efficiency, and reliability through **batching, caching, optimized data flows, retries, failure recovery, logging, and monitoring**.

Cura Technologies Inc Bangalore, IN

Software Engineer Dec 2019 - July 2021

- Built backend APIs and distributed services using **Python, Django, Flask, and PostgreSQL** for a healthtech platform serving **10+ hospitals and ~600 active users**.
- Led migration of backend services to **Kubernetes/EKS**, defining containerization, deployment, and scaling practices to improve reliability and operational consistency.
- Developed a **custom Kubernetes operator** to automate PostgreSQL provisioning, failover management, and lifecycle operations.
- Designed AWS infrastructure using **Terraform** and implemented rolling and blue-green deployments to improve scalability and release reliability.
- Built Flask-based report-generation microservices to automate clinical reporting and improve turnaround time for hospital operations.
- Built Flutter-based healthcare functionality, including a **DICOM interface**, and contributed to migration from Swift to Flutter across iOS, Android, and web.
- Worked directly with doctors, nurses, healthcare operations teams, and other stakeholders to translate clinical requirements into production improvements.

Tapchief (Relevel) Bangalore, IN

Product Engineer Oct 2018 - Oct 2019

- Built and shipped end-to-end product features across **Python, Node.js, PostgreSQL, Angular, and React Native** for a platform serving **~30K daily active users and ~100K daily API requests**, owning workflows from backend APIs through frontend integration and production delivery.
- Developed **multi-step onboarding and real-time communication workflows** using Angular, WebRTC, and Twilio, supporting **~80–100 audio/video calls per day** across web and mobile products.
- Built internal operational tooling and improved **debugging, deployment stability, and production reliability** across backend and frontend systems.

Education

Siliguri Institute of Technology Bachelor of Technology • 7.79

Computer Science and Engineering Aug 2013 - Jun 2018

Skills

Core Engineering: Distributed Systems, System Design, Backend Architecture, Microservices, API Design, Event-Driven Architecture

Languages & Frameworks: Go, Python, TypeScript, JavaScript, FastAPI, Django, Flask, Node.js, NestJS

AI/LLM Systems: AI Agents, Agentic RAG, Tool Calling, Prompt Engineering, Semantic Search, Text-to-SQL/NL-to-SQL, Text-to-API/NL-to-API, LLM Evaluation/RAGAS, Prompt Safety, PII Filtering

Frontend & Mobile: React, Angular, Flutter, React Native, WebRTC, Twilio

Cloud & Infrastructure: AWS, EKS, Lambda, S3, GCP, GKE, Cloudflare, Kubernetes, Docker, Terraform, CI/CD

Data & Storage: PostgreSQL, MySQL, MongoDB, Snowflake, BigQuery, Redis, Qdrant, Data Modeling, Vector Search, Embeddings

Messaging & Workflows: RabbitMQ, Queues, Pub/Sub, Asynchronous Processing, Workflow Orchestration, gRPC, Service-to-Service Communication

Reliability & Observability: Fault Tolerance, High Availability, Performance Optimization, Caching, Structured Logging, Distributed Tracing, Monitoring, Production Debugging

Leadership & Delivery: Technical Leadership, Mentoring, Stakeholder Communication, Requirement Gathering, Customer-Facing Engineering, User Feedback-Driven Iteration

Projects

Vidura — Self-hosted Agentic RAG System Feb 2024 – Oct 2024

- Built a production-grade agentic RAG system with retrieval, memory, and tool execution for contextual AI responses.
- Designed multi-step agent workflows with semantic search, document grounding, and self-hosted deployment support.

https://vidurabot.com/

Agent Orchestration Platform

- Built a full-stack AI agent orchestration platform with Next.js, TypeScript, FastAPI, PostgreSQL, Redis, and Python workers.
- Designed workflow execution, agent configuration, tool execution, Telegram webhook integration, message history, and runtime event mirroring.
- Implemented visual workflow templates, workflow runs, token/cost tracking, and OpenAI-backed agent execution with OpenClaw runtime integration.

https://github.com/ratnadeep007/agent-orchestration-platform

NL-to-SQL

- Built an NL-to-SQL system that converts natural-language analytics questions into safe SQL over structured datasets.
- Developed a Streamlit chat UI, Starlette/Uvicorn API, CLI, runtime schema discovery, SQL validation guardrails, and SQLite-backed demo database.
- Implemented OpenAI-based intent classification, schema-grounded SQL generation, SQL parsing/validation with sqlglot, caching, repair attempts, and automated tests.

https://github.com/ratnadeep007/nl-to-sql

SkyCUA — Computer Use Agent

- Built a local computer-use agent capable of interacting with system tools through containerized Docker environments.
- Designed execution workflows for task automation, tool orchestration, and controlled local agent operations.

https://github.com/ratnadeep007/skycua

GoClaw — Agent CLI & HTTP Service

- Built an agent system supporting tool execution, LLM interaction, persistent memory, and external integrations.
- Designed CLI and HTTP service interfaces for multi-step reasoning workflows and agent runtime experimentation.

https://github.com/ratnadeep007/goclaw